AI-Powered Exam Preparation Planner

Web Techniques Project

CRUD | MySQL Database | PHP Backend | AI API

First-Year Undergraduate · Web Techniques · Prof. Soufiane Hourri

Project Context — Why This App?

1

Students struggle to prioritize

When exams approach, students don't know which module to revise first. All subjects feel equally urgent.

2

Every student is different

Each student has different progress levels, understanding, and difficulty across modules.

3

AI can plan smarter

An AI API can analyze all this data and generate a personalized 1-week exam revision plan automatically.

Project Objective

Build a web application that does 3 things:

Manages Modules

Lets a student add, view, edit and delete their academic modules in a clean web interface.

Stores Data

Persists all module information in a MySQL database using PHP as the backend.

Uses AI

Sends module data to an AI API to generate a smart, personalized 7-day revision plan.

Main Features — What Can the App Do?

Add a Module

Create a new module with all its details.

Display Modules

View a list of all your modules.

View Module Details

See full information for one module.

Edit a Module

Update module info anytime.

Delete a Module

Remove a module you no longer need.

Generate AI Plan

Get a personalized 7-day revision plan.

CRUD Project + AI Integration

What is CRUD? — Applied to This Project

C Create

Example:

Add a new module (e.g. "Mathematics") to the database.

R Read

Example:

Display all your modules on the main page.

U Update

Example:

Edit module difficulty or change the exam date.

D Delete

Example:

Remove a module you no longer need.

Module Information — Required Fields

Each module stores 7 pieces of information that help the AI prioritize revision:

Module Name

e.g. "Mathematics", "Web Techniques"

Current Progress

0 – 100 %

Teacher

Name of the module's teacher

Understanding Level

Low / Medium / High

Difficulty

Easy / Medium / Hard

Exam Date

Date of the upcoming exam

Career Importance

Low / Medium / High

How the AI Is Useful

What goes IN

- Module difficulty (Easy/Medium/Hard)
- Career importance (Low/Medium/High)
- Current progress (%)
- Understanding level
- Exam date (urgency)

AI API

What comes OUT

- ✓ A smart 7-day revision plan
- ✓ Harder modules scheduled first
- ✓ Urgent exams prioritized
- ✓ Study time adapted to progress
- ✓ Fully personalized per student

AI is not used randomly — it solves a real problem for every student.

Example of AI-Generated Revision Plan

Sample output for a student with 5 modules:

Day 1

Web Techniques

2 hours

Mathematics

1 hour

Algorithms

1 hour

Day 2

Database

2 hours

English

1 hour

Day 3

Mathematics

2 hours

Web Techniques

1 hour

Day 4

Algorithms

2 hours

Database

1 hour

Difficult and urgent modules are always scheduled first in the week.

Technologies Required

HTML5

Structures the web pages and forms

CSS3

Designs and styles the interface

PHP

Backend logic — connects forms to database

MySQL

Stores all module data in tables

WampServer XAMPP

Runs Apache + PHP + MySQL locally

What is Localhost?

Localhost = your own computer as a server

- You don't need to upload files to the internet.
- WampServer or XAMPP runs a local web server (Apache) on your PC.
- PHP and MySQL work exactly the same as on a real server.
- Once ready, you can move your project online easily.

```
http://localhost/exam_planner
```

Your laptop IS the server during development!

Development Environment — WampServer / XAMPP

Install WampServer or XAMPP to get these three tools in one package:

Apache

Web Server

- Serves your PHP pages to the browser
- Runs on port 80 by default
- Start it with one click in WampServer

MySQL

Database

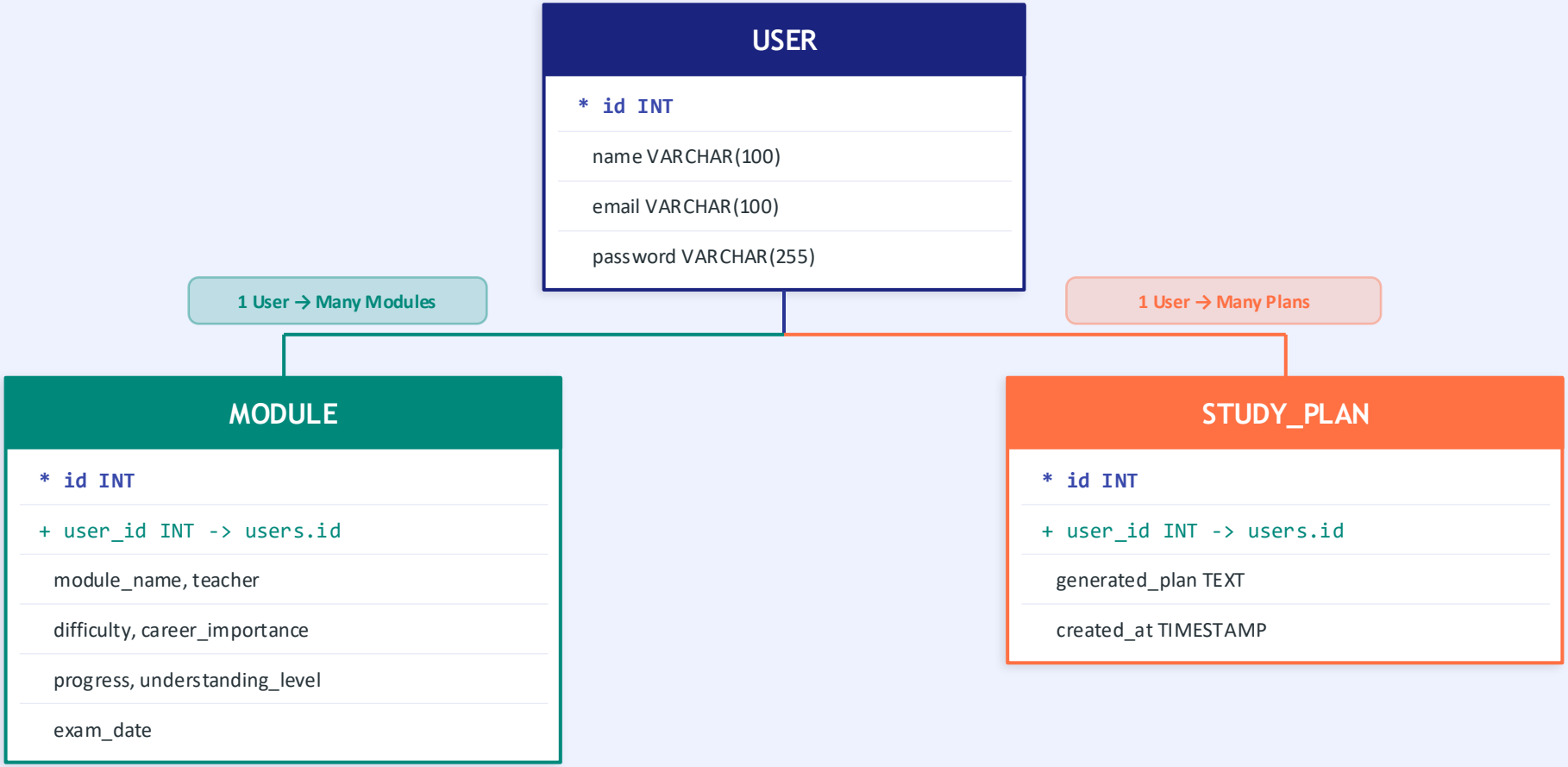
- Stores all module data in tables
- Manage via phpMyAdmin in browser
- Used by PHP to read/write data

PHP

Backend Language

- Processes form submissions
- Connects to MySQL database
- Generates HTML pages dynamically

Entity Relationship Diagram



Database Tables — Structure at a Glance

users

Column	Type	Notes
id	INT	AUTO_INCREMENT PRIMARY KEY
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	NOT NULL UNIQUE
password	VARCHAR(255)	NOT NULL

study_plans

Column	Type / Constraint
id	INT — AUTO_INCREMENT PK
user_id	INT — FK → users.id
generated_plan	TEXT
created_at	TIMESTAMP DEFAULT NOW()

modules

Column	Type / Constraint
id	INT — AUTO_INCREMENT PK
user_id	INT — FK → users.id
module_name	VARCHAR(150) NOT NULL
teacher	VARCHAR(100)
difficulty	ENUM('Easy','Medium','Hard')
career_importance	ENUM('Low','Medium','High')
progress	INT (0 to 100)
understanding_level	ENUM('Low','Medium','High')
exam_date	DATE NOT NULL
created_at	TIMESTAMP DEFAULT NOW()

8-Week Development Plan

1 Week 1

Environment Setup

Install WampServer/XAMPP. Create the database & tables with phpMyAdmin.

2 Week 2

Database Design

Verify tables & relationships. Insert test data. Explore phpMyAdmin.

3 Week 3

HTML & CSS Interface

Build page layout, navigation, and module forms with HTML5 and CSS3.

4 Week 4

PHP — Read & Display

Connect PHP to MySQL. Fetch and display the list of modules.

5 Week 5

Create & Delete

Implement the Add Module form (Create) and the Delete functionality.

6 Week 6

Update & Details Page

Add Edit form (Update) and a module detail view page.

7 Week 7

AI API Integration

Connect to an AI API. Send module data and display the generated plan.

8 Week 8

Testing & Polish

Test all features, fix bugs, improve UI, prepare for presentation.